

AMMAR ALI

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EDUCATION

New York University

Master of Science in Computer Science

Expected May 2028

Brooklyn, NY

Stony Brook University

Bachelor of Science in Computer Science, Specialization in Artificial Intelligence and Data Science | GPA: 3.91 *Stony Brook, NY*

May 2026

Courses: Software Engineering, System Fundamentals, Data Structures, Artificial Intelligence, Data Science, Scripting Languages

Honors: Presidential Scholarship, Outstanding Academic Achievement Award, Dean's List (all semesters)

SKILLS

Languages: Python, Java, C, JavaScript/TypeScript, HTML/CSS, C++, SQL, OCaml, R

Frameworks & Tools: React, Node.js, Express.js, MongoDB, Socket.io, Jest, Git, Linux/Unix, Jira

Cloud & Deployment: AWS (S3, App Runner, ECR), Docker, CI/CD Pipelines, Firebase

AI/ML: PyTorch, RLHF, Claude API, PPO, Scikit-learn, Pandas, NumPy, HuggingFace, Replicate

EXPERIENCE

Automotive Ethics VIP Lab, Stony Brook University

Lead Software & Machine Learning Engineer

Aug 2024 – Present

Stony Brook, NY

- Architected a production ML system encoding Kantian and Utilitarian ethical frameworks into autonomous vehicle decision-making, training policy, value, and reward networks (~12K parameters) via PyTorch with Bradley-Terry preference learning over 200+ labeled human trajectories.
- Led a team of 7 engineers through full development lifecycle including PPO training, Generalized Advantage Estimation, preference pipeline, and CARLA simulator API integration; managing sprint planning and delivery milestones via Jira.
- Pivoted project scope from hardware to scalable ML/simulation, reducing projected delivery time by 200%; submitted conference abstracts and preparing publication for AI ethics venues.

Suffolk County Department of Information Technology

Software & Web Development Intern

Jun 2025 – Jul 2025

Hauppauge, NY

- Designed and developed 3 county initiative websites using HTML, CSS, and JavaScript for distinct county initiatives; maintained and updated 50+ web pages via the county CMS to enhance usability and keep content current.
- Collaborated on the implementation and testing of AI-powered tools such as a chatbot (built on Google Web Crawler infrastructure) intended for integration into the county's official website; evaluated the chatbot's responsiveness to follow-up queries and refined key performance parameters to improve user comprehension and interaction by 50%.
- Executed cross-server QA testing to ensure site consistency and performance across testing and production environments.

PROJECTS

Draftroom Live Fantasy Baseball Platform (50+ users) | TypeScript, React, Node.js, Express.js, MongoDB, AWS, Socket.io, Docker

- Architected a dual-product system: a publicly documented REST API (Amethyst API, with open key registration) delivering context-aware player valuations, and a full-stack web app built on a shared TypeScript monorepo using pnpm workspaces.
- Engineered a context-aware valuation engine that caches daily MLB Stats API calls and dynamically re-prices players mid-draft based on league-wide roster state, minimizing redundant API calls while keeping valuations accurate throughout live drafts.
- Built an AI-driven mock draft simulator where 11 autonomous AI teams independently manage budgets and roster needs, placing competitive bids against live valuations, giving users realistic auction draft preparation.
- Owned end-to-end delivery including React frontend, backend API, MongoDB Atlas multi-league data isolation, JWT/bcrypt authentication, and AWS cloud deployment (S3, App Runner, Docker/ECR); contributed to CI/CD pipeline configuration.

Stock Exchange Server | C, POSIX Threads, Sockets

- Implemented a concurrent stock exchange server supporting simultaneous multi-client trading via POSIX threads, with buy/sell order matching, bid-ask spread tracking, and real-time ticker broadcast to all connected traders over TCP.
- Designed thread-safe data structures using mutexes (including recursive mutexes for re-entrant locking), semaphores, and reference counting to eliminate race conditions and memory leaks under concurrent load.

ReminisceAI Generative AI for Dementia Care (In Development) | Claude API, Flux Dev LoRA, Replicate, AWS S3, MongoDB,

- Co-founding an AI startup building a reminiscence therapy platform for dementia patients, using a multi-modal generative pipeline to transform patient memories (text prompts, photos, music) into enriched, personalized VR sessions.
- Architecting an end-to-end memory enrichment pipeline leveraging Claude API for narrative generation and Flux Dev LoRA (via Replicate) for personalized image synthesis; storing and serving memory assets via AWS S3 and MongoDB.